Statistical Methods in Education Research
Vanderbilt University
Prof. Sean P. Corcoran

Due November 17, 2026
58 total points

Problem Set 10

Instructions: Answer the following questions in their entirety in a separate document. Submit your completed problem set as a PDF document via email to `sean.corcoran@vanderbilt.edu`. Use your last name and problem set number as the filename. Working together is encouraged, but it is expected that all submitted work be that of the individual student.

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1. The Zagat Restaurant Guides rate each restaurant on a 30-point scale for food, decor, service, and cost. The *zagat.dta* file on Github shows 2007 ratings for Italian restaurants in Boston, London, and New York. (**12 points—4 each**)

```
use https://github.com/spcorcor18/LPO-8800/raw/main/data/zagat.dta
```

   (a) Create a scatterplot that shows the relationship between the food quality rating (on the vertical axis) and the cost rating (on the horizontal axis). Find and interpret the correlation coefficient between these variables.

   (b) Now create the same scatterplot but use separate colors to differentiate the data points for each city. (Hint: the `twoway` command allows you to plot multiple scatterplots on the same graph, using different colors. In the drop-down menus, use Graphics → Twoway). Find and interpret the correlation coefficients separately by city.

   (c) Contrast your findings in parts (a) and (b). Why do they appear to tell different stories? How is this a cautionary tale for the interpretation of correlation coefficients?

2. Read the dataset called *states.dta* on Github. This file contains some education-related data from the early 2000s for the 50 states plus D.C. (**18 points—3 each**)

```
use https://github.com/spcorcor18/LPO-8800/raw/main/data/states.dta
```

   (a) Create a scatterplot showing the relationship between the average teacher's salary (*teachpay*) and education expenditure per pupil (*educexpe*). Label the data points in your scatter plot with the state names. (Hint: when creating the graph using Graphics → Twoway, you have the ability to create marker labels under "marker properties").

(b) Identify a state that appears to be an outlier with respect to the general relationship between teacher salaries and expenditure. In what way is this state unusual? Re-calculate the correlation coefficient without this state and report what you find.

(c) Create a scatterplot matrix showing the relationships between the following variables: average educational expenditure per pupil (*educexpe*), student teacher ratio (*stuteach*), the average verbal SAT score for the state (*satv*), the average teacher salary (*teachpay*), and the high school graduation rate (*gradrate*). Consider average educational expenditure. Do the associations between this and each of the other variables appear linear?

(d) Create a correlation matrix between all pairs of the variables above. Which correlations with average educational expenditure are strong and which are less so? Which has the strongest association?

(e) How many of the correlations in part (d) are statistically significant? Which, if any, are not?

(f) Apply the Bonferroni correction to the correlation matrix in part (d). How does this change your answer, if at all, to part (e)? Explain in words why one might want to do this.

3. A state education agency is interested in whether school accountability ratings are associated with the type of community in which a school is located. The table below presents accountability ratings for a sample of 600 public schools. **(12 points)**

School locale	**High rating**	**Medium rating**	**Low rating**	**Total**
Urban	80	100	70	250
Suburban	80	85	35	200
Rural	40	55	55	150
Total	200	240	160	600

(a) Calculate the percentage of schools receiving a *low* accountability rating within each of the three locale types. Describe the differences you observe. **(3 points)**

(b) State the null and alternative hypotheses for a chi-squared test of independence. **(2 points)**

(c) Calculate the expected frequencies for urban schools with low ratings, suburban schools with high ratings, and rural schools with medium ratings under the null hypothesis. **(3 points)**

(d) Conduct a chi-squared test of independence using all nine cells. Report the test statistic, degrees of freedom, and p-value. Interpret your findings at the 5% significance level. I recommend you do this in Stata using the table calculator. **(4 points)**

4. Begin with the "toy" dataset created by the following Stata syntax. For each pair of variables (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), and (x_5, y_5), create a scatterplot and calculate both the Pearson and Spearman correlation coefficients. (Put the y values on the y-axis). For each pair, explain why and how the two correlations differ, or why they do not. In which case(s) are the Spearman correlations particularly useful? (**5 points**)

```
set seed 8800
set obs 250

* 1. Linear relationship with random noise
gen x1 = runiform(0,10)
gen y1 = 15 + 2*x1 + rnormal(0,5)

* 2. Nonlinear, monotonic relationship
gen x2 = runiform(0,4)
gen y2 = 4*exp(.8*x2) + rnormal(0,5)

* 3. U-shaped relationship
gen x3 = runiform(-3,3)
gen y3 = 4*x3^2 + rnormal(0,3)

* 4. Linear relationship with an unusual observation
gen x4 = runiform(0,10)
gen y4 = 10 + 2*x4 + rnormal(0,4)
replace x4 = 22 in 1
replace y4 = -100 in 1

* 5. Independent variables
gen x5 = runiform(0,10)
gen y5 = rnormal(20,6)
```

5. Begin with the dataset *card.dta* on Github. These data come from a study by David Card (1995) that estimated the earnings returns to additional years of education. (**11 points**)

```
use https://github.com/spcorcor18/LPO-8800/raw/main/data/card.dta
```

(a) Using the full dataset, calculate the correlation between years of education (*educ*) and the individuals log annual earnings (*lwage*). Inspect the scatterplot to determine whether the correlation coefficient is appropriate for these variables. For the purposes of parts (b)-(c), consider these data the population. (**3 points**)

(b) Now run the following syntax. Based on what you have done in previous assignments, explain in words what this code is doing. (**3 points**)

```
bootstrap r(rho), size(50) reps(100) saving(results): corr educ lwage
```

(c) Using the resulting dataset in part (b), produce a histogram for "rho" and a set of descriptive statistics. What is the histogram showing you? Between what values did the correlation coefficient fall 95% of the time? (**5 points**)